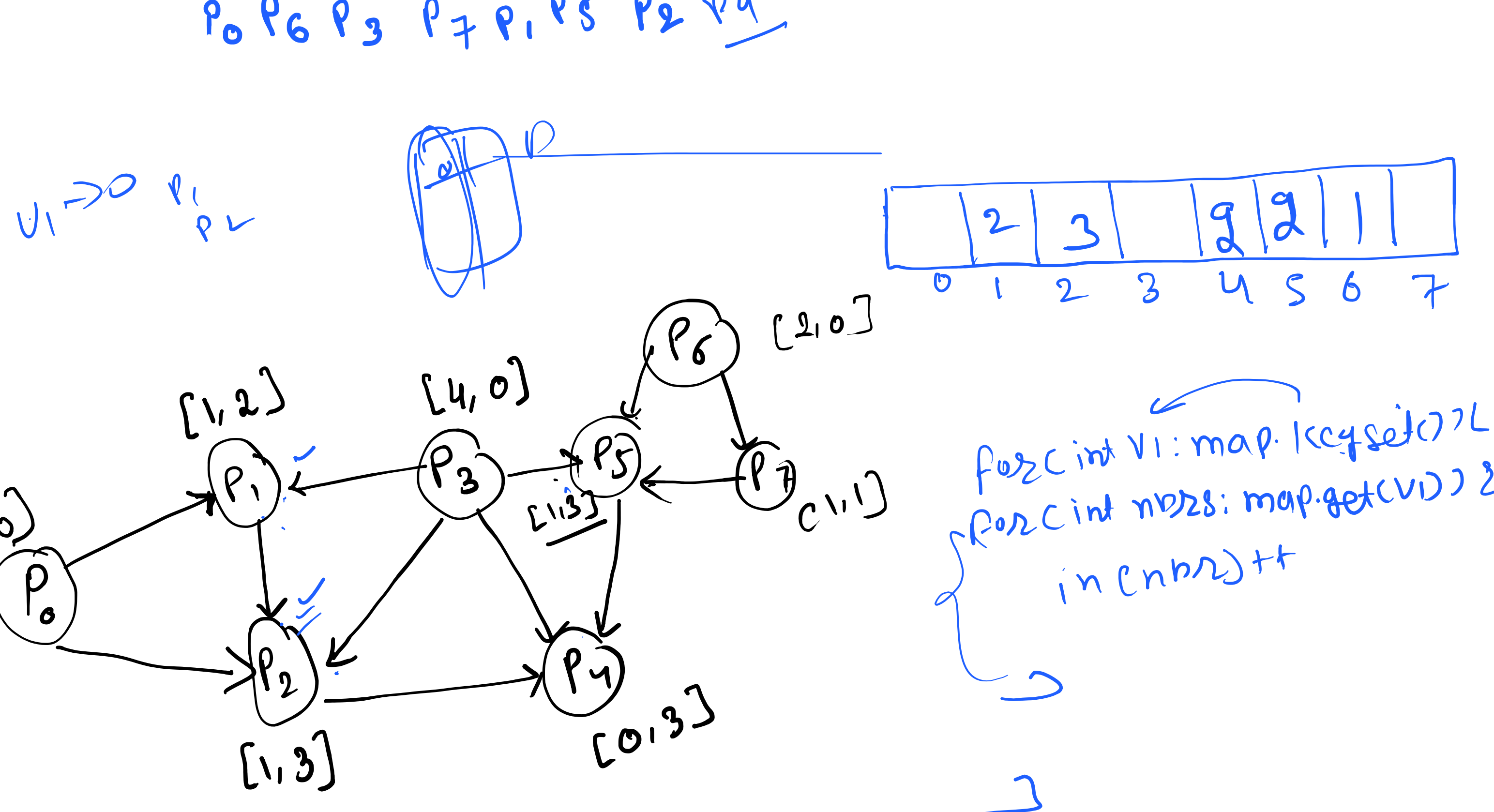
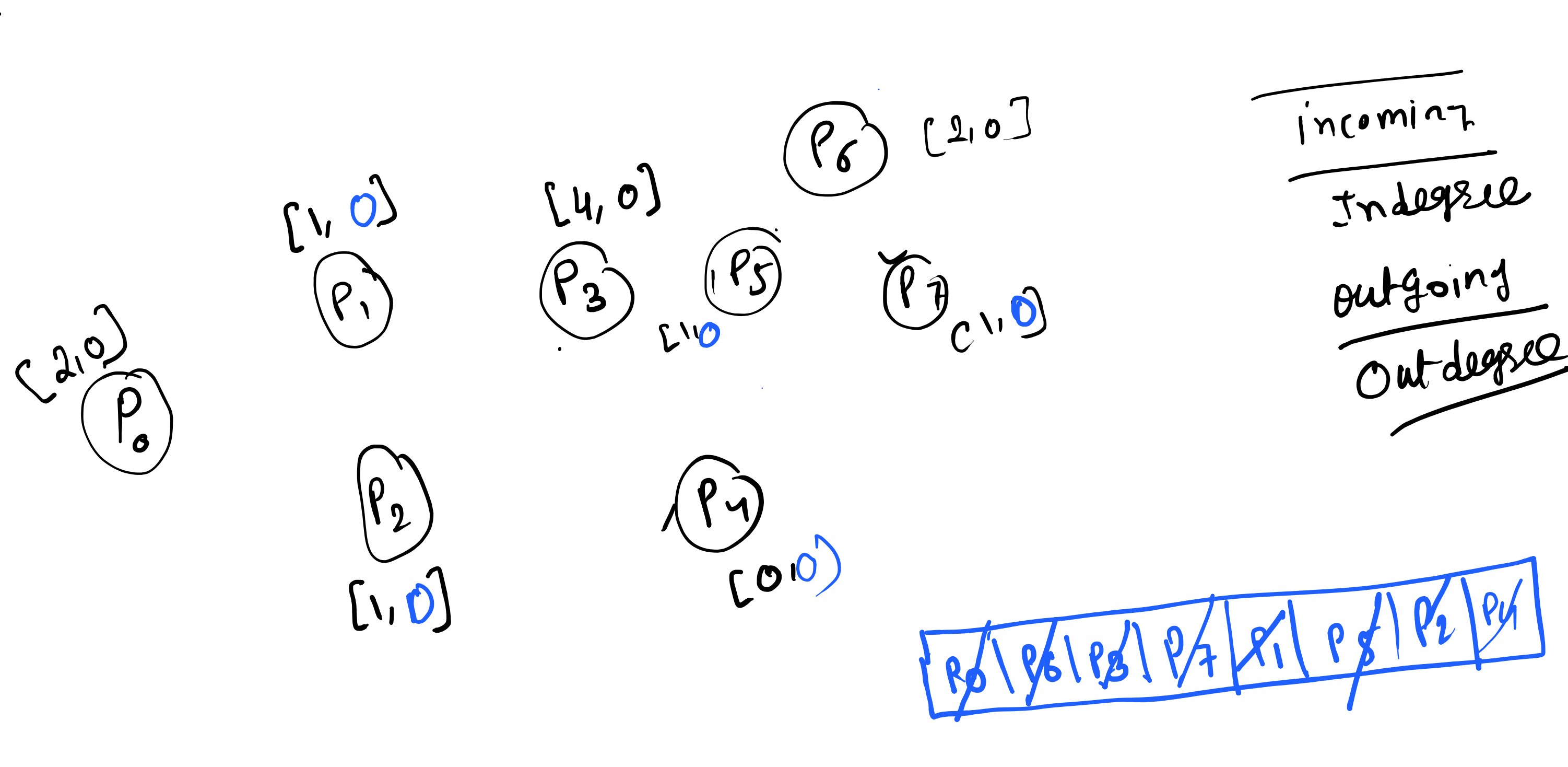
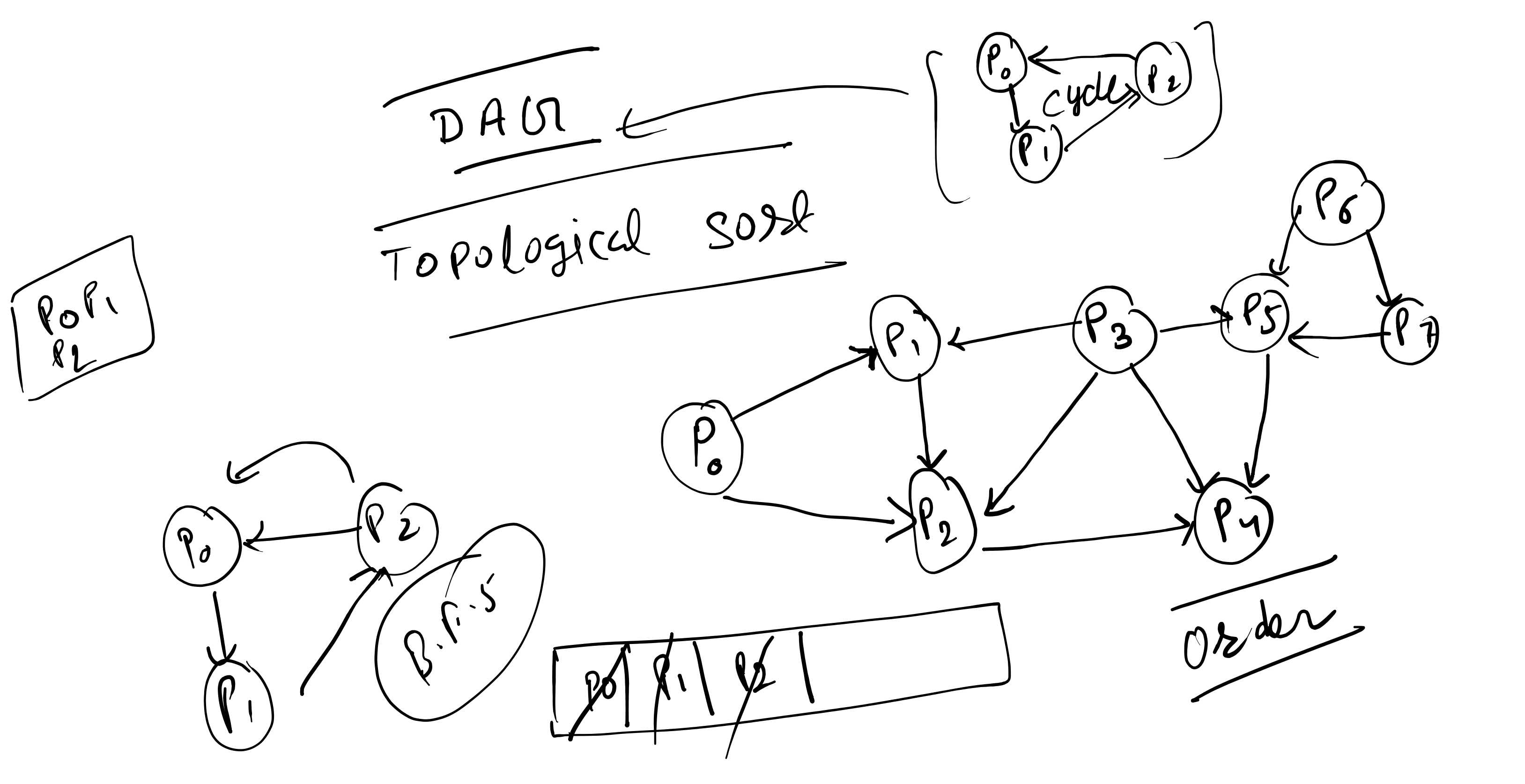
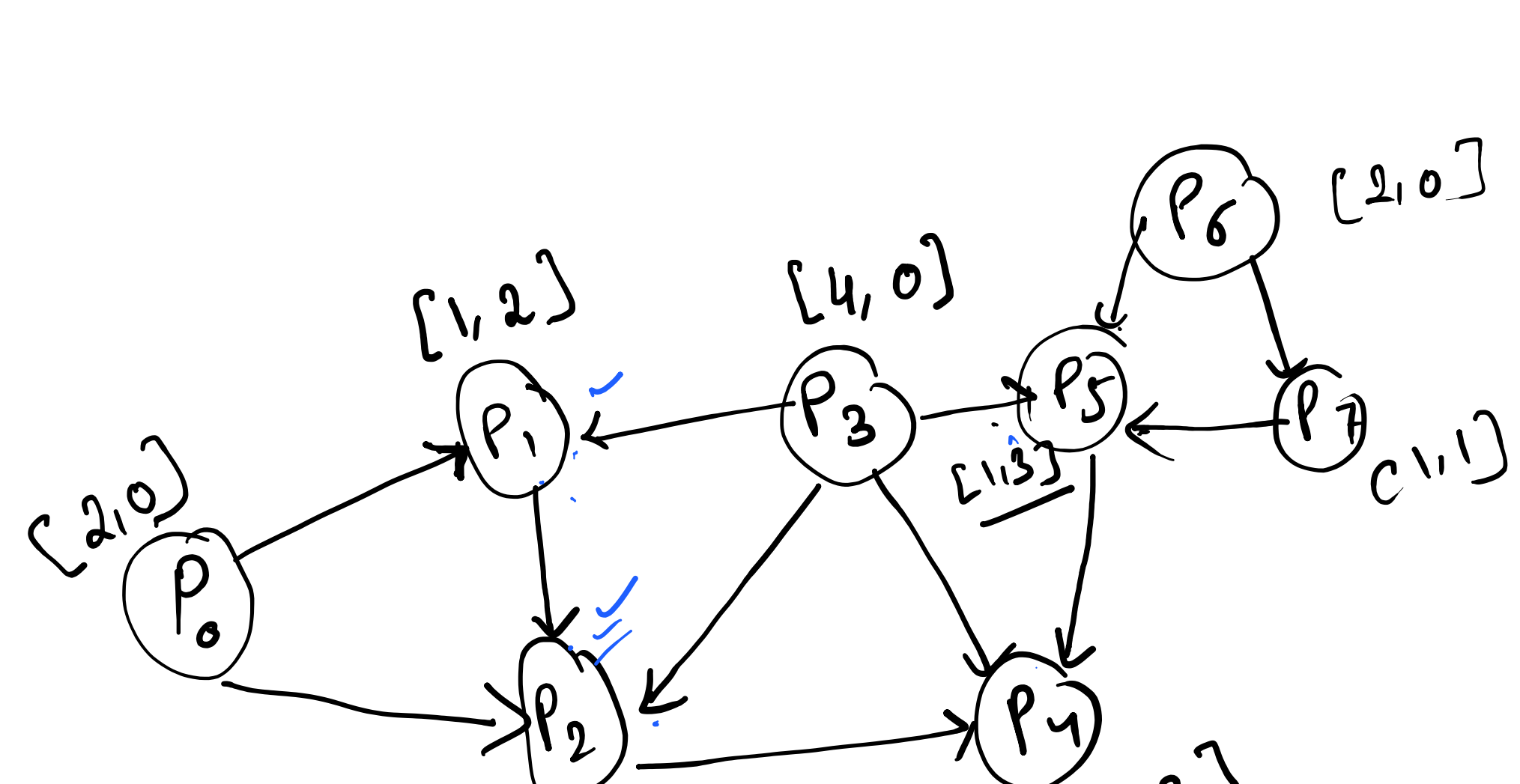
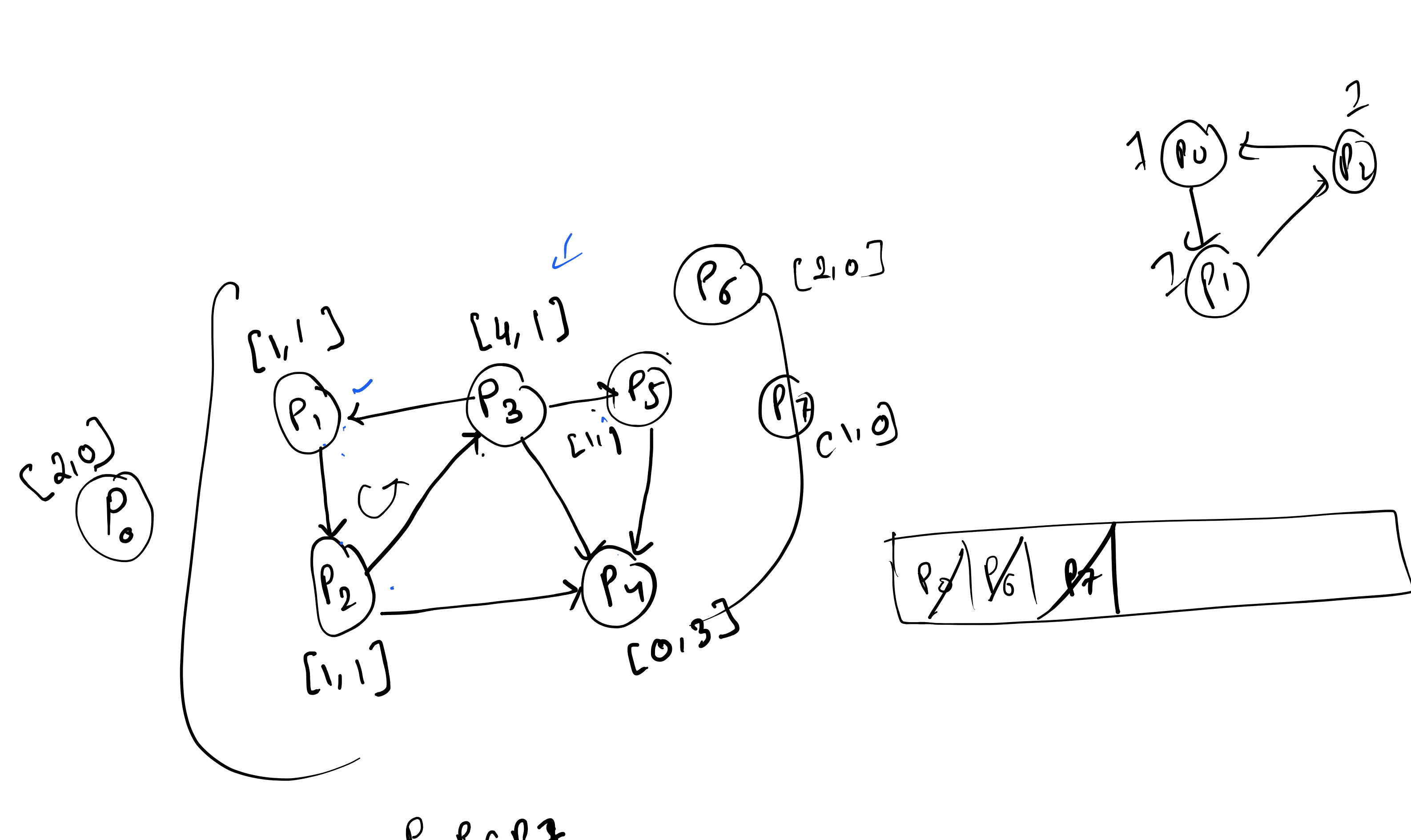
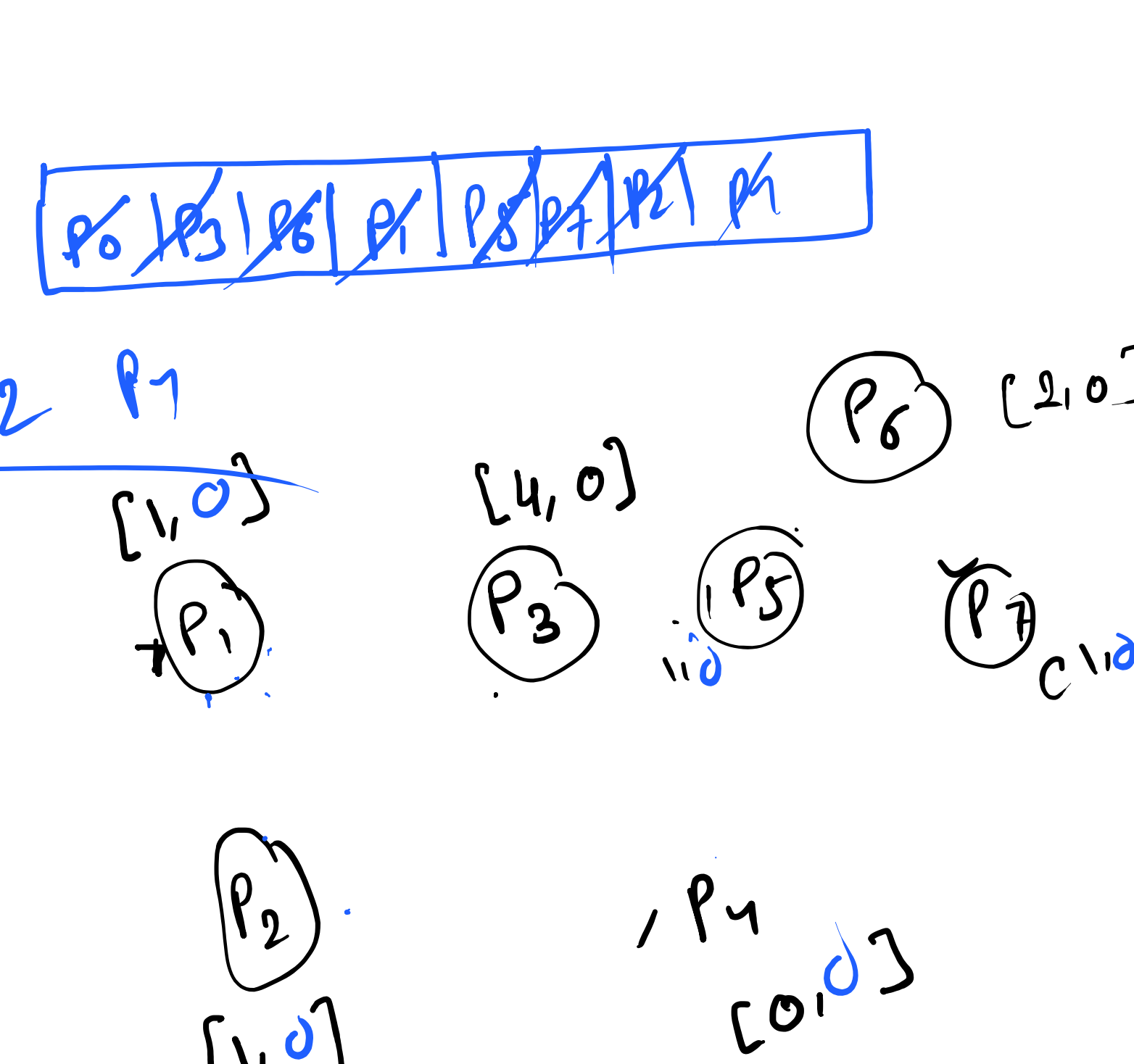




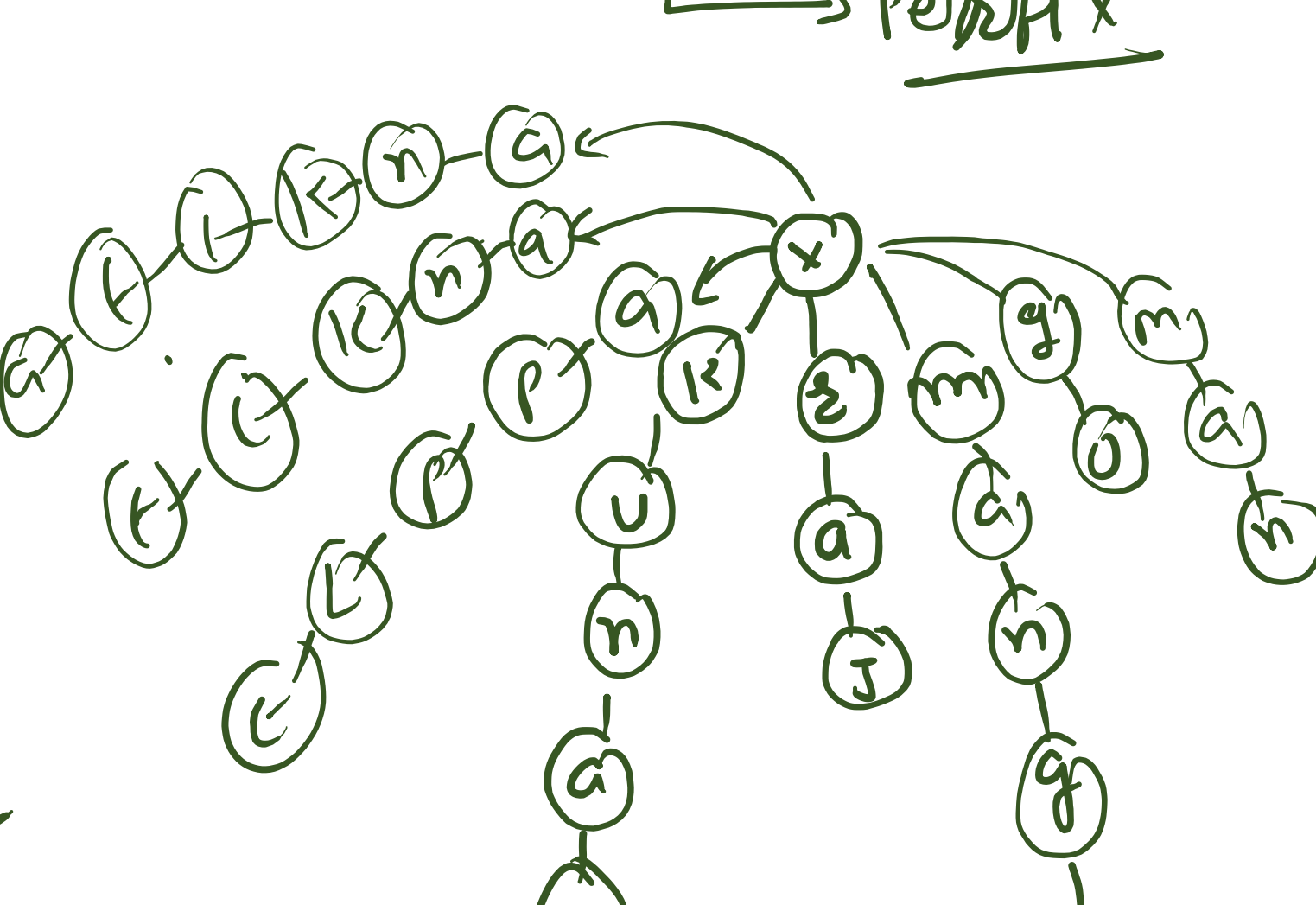
```
public void topological_sort() {  
    int[] in = Indegree();  
    Queue<Integer> q = new LinkedList<>();  
    for (int i = 0; i < in.length; i++) {  
        if (in[i] == 0) {  
            q.add(i);  
        }  
    }  
    while (!q.isEmpty()) {  
        int r = q.poll();  
        System.out.print(r + " ");  
        for (int nbrs : map.get(r)) {  
            in[nbrs]--;  
            if (in[nbrs] == 0) {  
                q.add(nbrs);  
            }  
        }  
    }  
    System.out.println();  
}
```



Trie

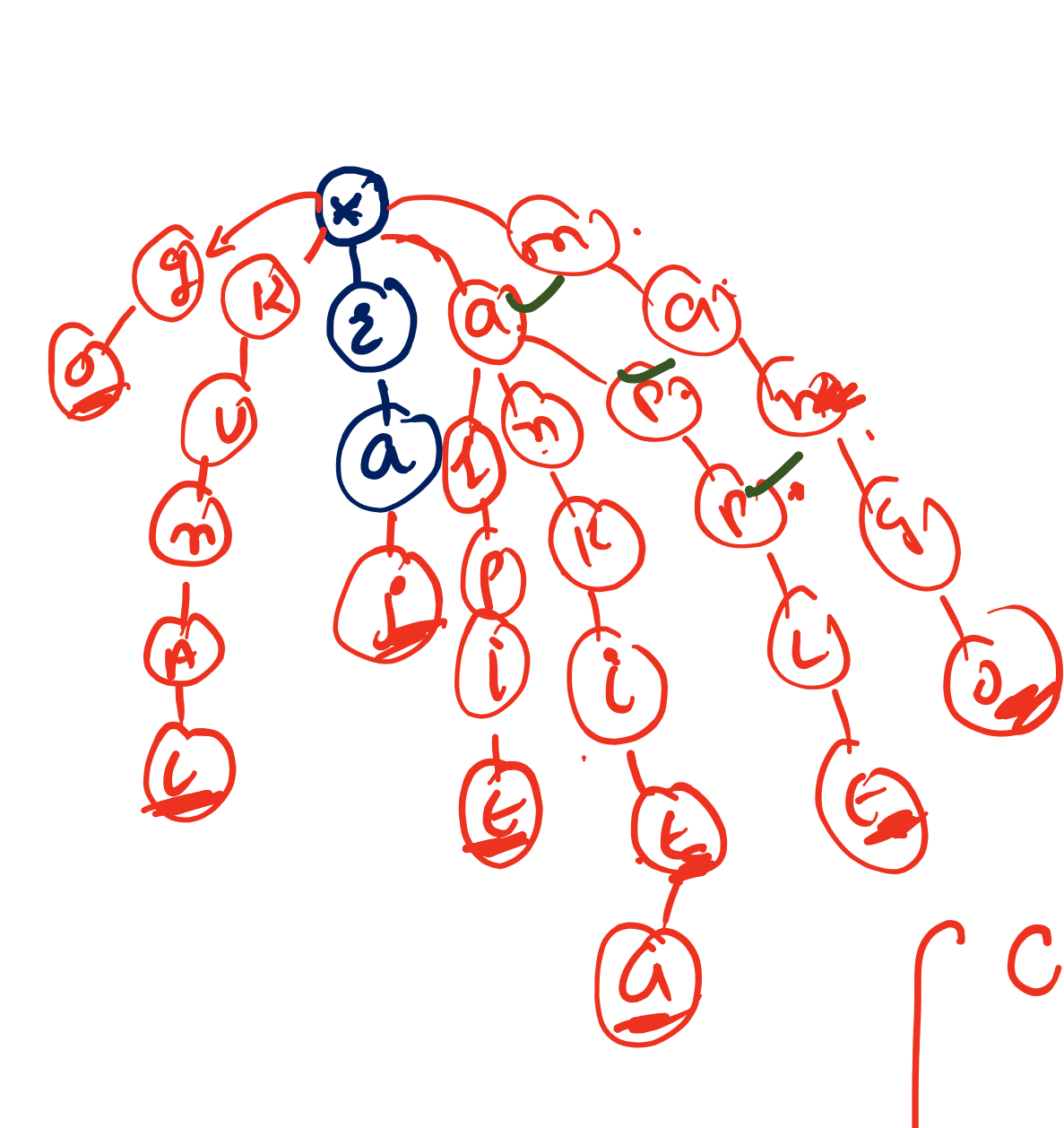
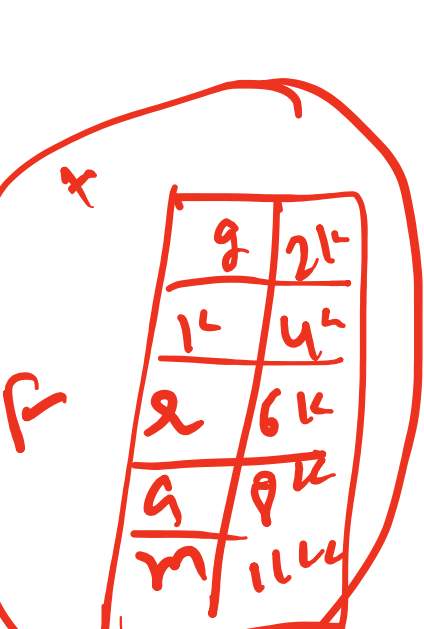
Perfix

- i) Raj
- ii) Kunal
- iii) mango
- iv) Ankit
- v) go
- vi) Apple
- vii) ankita
- viii) man



Who is

- i) Raj
- ii) Kunal
- iii) mango
- iv) Ankit
- v) go
- vi) Apple
- vii) ankita
- viii) man

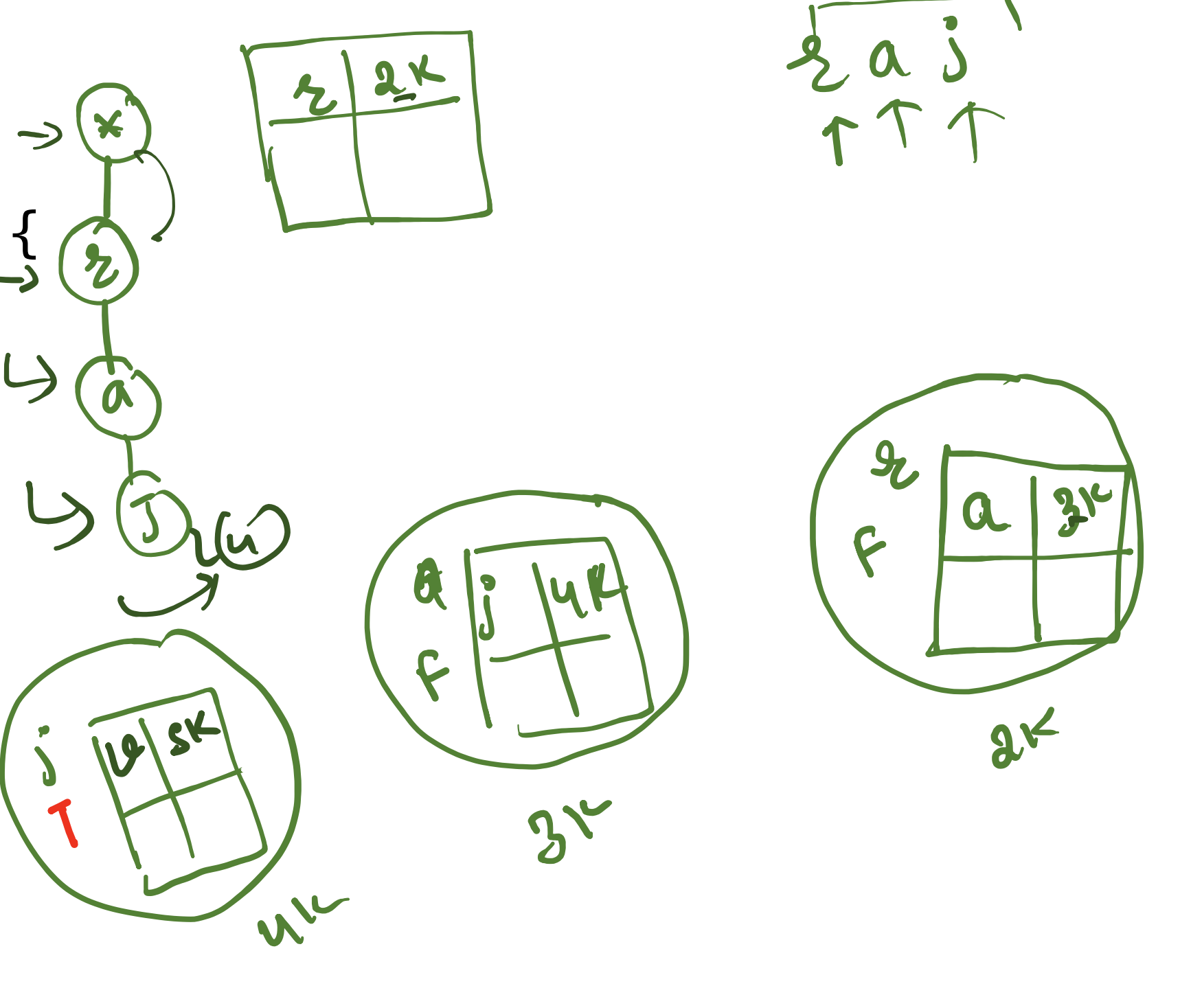


mango go

app

Class Node {
 char ch
 HashMap<char, Node> child
 boolean isTerminal
}

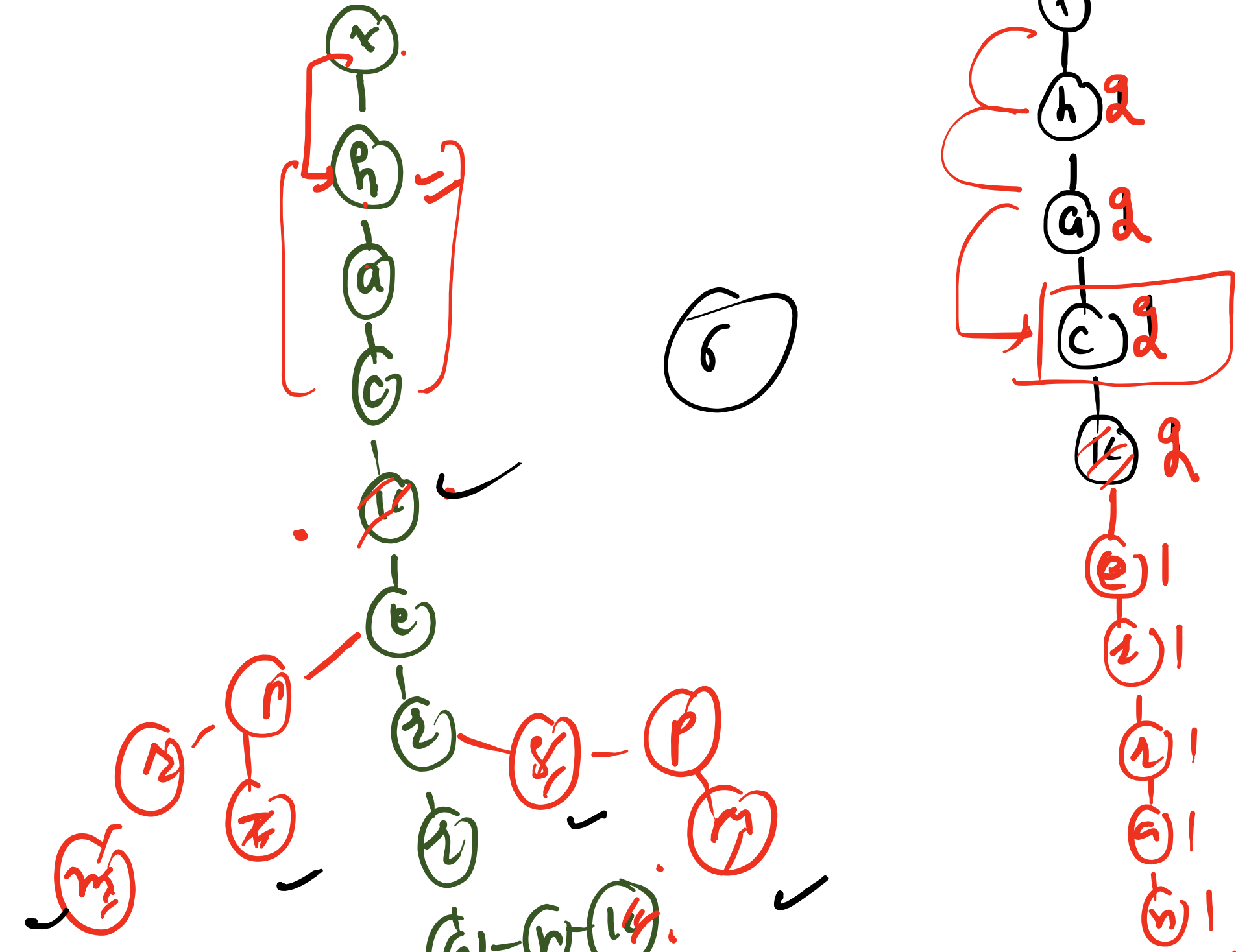
```
public void insert(String word) {  
    Node curr=root;  
    for (int i = 0; i < word.length(); i++) {  
        char ch = word.charAt(i);  
        if (curr.containsKey(ch)) {  
            curr = curr.child.get(ch);  
        }  
        else {  
            Node nn = new Node(ch);  
            curr.child.put(ch, nn);  
            curr=nn;  
        }  
    }  
    curr.isTerminal=true;  
}
```



4
add hack
add hackerrank
find hac
find hak

Sample Output

2
0



001111

```
int[] arr = { 2, 1, 1, 0, 1, 2, 5, 4, 0, 2, 8, 7, 9, 2, 6, 1, 9};
```

